

Treasury Register

AI walkthrough

Every AI feature we shipped, what problem each one solves, what value it adds, and how each one stays safe · book dated 3 September 2026

This walkthrough is a feature-by-feature tour of the AI layer. It assumes you have read the base walkthrough, which covers the app's controls before any AI is added — the six checks, the queue, the deal lifecycle, the two exposures. The features here layer on top of that foundation without changing it.

The one design principle

Every AI feature in the register follows the same shape: **numbers deterministic, prose AI, six checks always gate the book**. Every figure a treasurer sees in an AI-produced card is computed by the same code that computes today's live figures — the exposure calculator, the check engine, the accrual service. The AI's role is to write the paragraph that reads those figures, rank the options, name the trade-off. It does not decide any number.

That means the AI could go wrong in exactly one way: it could recommend the wrong option. It cannot invent a rate, exceed a limit, or write a deal. And if the model is unavailable, every feature falls back to a deterministic paragraph built from the same figures — slower to read, but never wrong.

Three signals every AI-produced element carries

- **The sparkle icon** — the AI-produced card always leads with a small ■ mark, so a reader learns to distinguish AI-produced content from deterministic content at a glance.
- **A reasoning trace** — before any AI paragraph is shown, the six steps the system is actually running tick through in the domain vocabulary ("Taking a snapshot of the current book", "Applying the rating action to a scratch copy"). Each step corresponds to real server-side work, so the trace is honest rather than theatre.
- **Real-time text** — the paragraph types itself out character by character with a blinking caret, and a *drafted just now* marker sits beside it. Both signal that the text is being generated for this call, not a canned caption.

Feature 01 · Advisory — ranked candidates for tomorrow's book

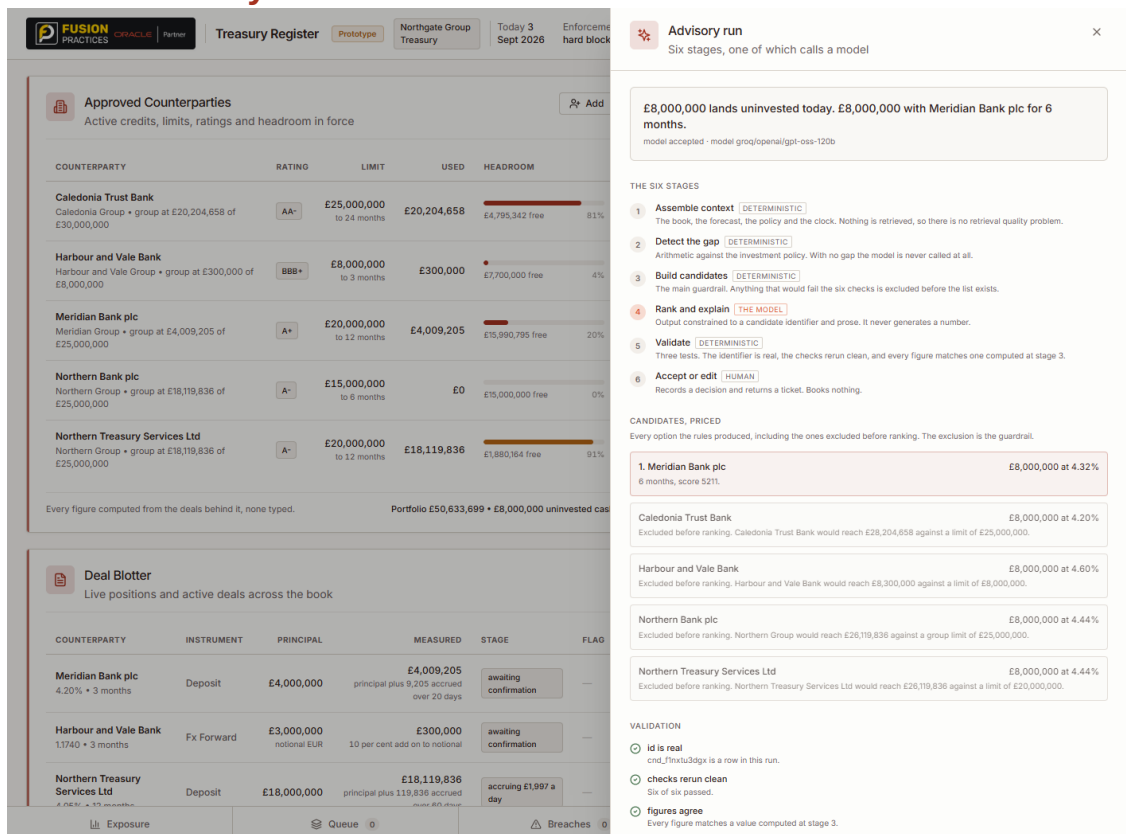


Figure 1. Advisory — ranked candidates for tomorrow's book.

What it solves

A treasurer starts every day facing the same question: where to place today's uninvested cash. The book has room in several counterparties; a spreadsheet lookup can find them, but ranking them by return, credit, tenor, and concentration is a judgment the treasurer usually does in their head. Under time pressure the choice defaults to whoever is easiest, not whoever is best.

What it adds

- Runs on a schedule, so the ranked list is waiting when the treasurer sits down.
- Only ever ranks candidates that already pass the six checks. Every option offered is bookable as-is.
- Records a full run in six labelled stages so the reasoning is auditable, not just the result.
- Load into the ticket sends the chosen deal through the ordinary control loop — the six checks re-run, the approver is named, and the deal is written by the same path as anything typed by hand.

How it stays safe

The model chooses between options already priced and gated by deterministic code. If it returns something malformed, the run falls back to a weighted score and the panel says so.

Feature 02 · What if the rating changes

The screenshot displays the Treasury Register interface. The main panel shows a table of 'Approved Counterparties' with columns for Counterparty, Rating, Limit, Used, and Headroom. The table lists five counterparties: Caledonia Trust Bank, Harbour and Vale Bank, Meridian Bank plc, Northern Bank plc, and Northern Treasury Services Ltd. Each entry includes a rating (e.g., AA-, BBB+, A+, A-), a limit (e.g., £25,000,000), and a used amount (e.g., £20,204,658). A progress bar indicates the percentage of the limit used (e.g., 81% for Caledonia Trust Bank).

Below the table is a 'Deal Blotter' section showing live positions and active deals. It includes columns for Counterparty, Instrument, Principal, Measured, Stage, and Flag. An example entry for Meridian Bank plc shows a deposit of £4,000,000 with a measured amount of £4,009,205.

On the right, a 'Ratings and policy' modal is open. It allows users to record a rating action by selecting a counterparty (Caledonia Trust Bank) and a new rating (BBB). The modal includes a 'What would this do?' section that provides a summary of the impact: 'Caledonia Trust Bank moves from AA- to BBB, tightening its limit to £5,000,000 (was £25,000,000). Two positions become breaches, raising open breaches on the book from 0 to 2. The treasurer must now monitor the higher breach count and assess whether additional credit controls are required.' The modal also features a 'What would this do?' button and an 'Apply, and re-test the book' button.

Figure 2. What if the rating changes.

What it solves

A rating action is expensive and rare, so treasurers rarely stress-test for it — which means when one lands, the answer "what does this do to our book?" is a spreadsheet exercise that takes half a morning. The register now answers it in seconds, before touching anything.

What it adds

- Pick a counterparty and a new rating; the register applies the rating action to a scratch copy of the book and reports back on limit, positions and breaches.
- The AI paragraph reads the deterministic before-and-after into a treasurer's language: "Caledonia's limit tightens from £25m to £5m, two positions become breaches, consider tightening usage."
- Preview always precedes action. The **What would this do?** button sits beside **Apply, and re-test the book**, and nothing changes in the book until the second button is pressed.

How it stays safe

The scenario runs inside a database savepoint that is rolled back at the end. Nothing about it survives; the book is not touched.

Feature 03 · What if the concentration cap moved

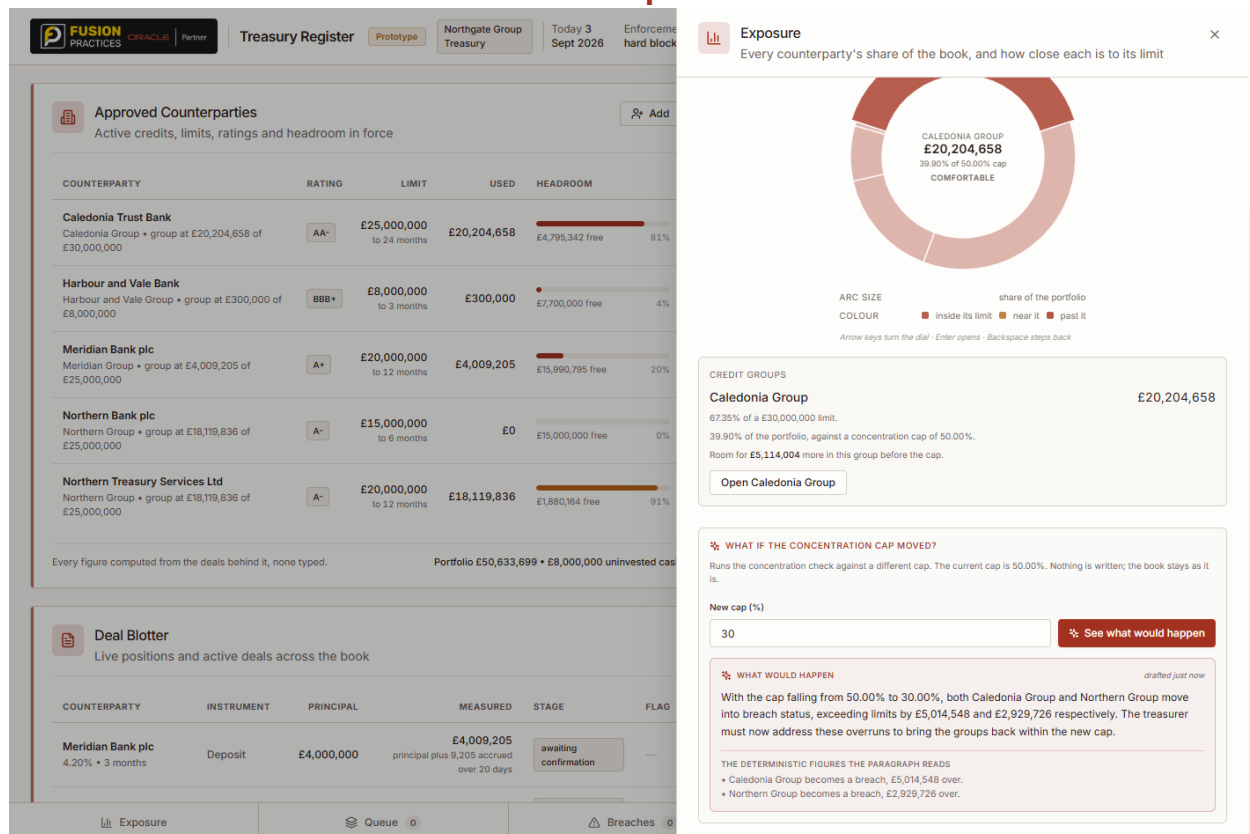


Figure 3. What if the concentration cap moved.

What it solves

Concentration policy sits somewhere between credit and risk, and a change to it usually starts a debate with no numbers on the table. "If we moved the cap to 40, how much would we be over?" — the answer needs the whole book plus the arithmetic of the concentration check, and nobody wants to run that in a meeting.

What it adds

- Enter a new cap percentage. The concentration check runs against it and every group that would cross the line is named, with the pence figure it would be over.
- The AI paragraph turns the result into an executive line: "Two groups become breaches, Caledonia is £5m over and Northern is £3m over."
- No writes at all. The scenario is pure calculation over the current state.

How it stays safe

Read-only. No savepoint, no writes, no database changes. Just the same concentration check the live gate runs, evaluated against a different threshold.

Feature 04 · What if a deal is not rolled at maturity

Approved Counterparties
Active credits, limits, ratings and headroom in force

COUNTERPARTY	RATING	LIMIT	USED	HEADROOM
Caledonia Trust Bank Caledonia Group • group at £20,204,658 of £30,000,000	AA-	£25,000,000 to 24 months	£20,204,658	£4,795,342 free 81%
Harbour and Vale Bank Harbour and Vale Group • group at £300,000 of £8,000,000	BBB+	£8,000,000 to 3 months	£300,000	£7,700,000 free 4%
Meridian Bank plc Meridian Group • group at £4,009,205 of £25,000,000	A+	£20,000,000 to 12 months	£4,009,205	£15,990,795 free 20%
Northern Bank plc Northern Group • group at £18,119,836 of £25,000,000	A-	£15,000,000 to 6 months	£0	£15,000,000 free 0%
Northern Treasury Services Ltd Northern Group • group at £18,119,836 of £25,000,000	A-	£20,000,000 to 12 months	£18,119,836	£1,880,164 free 91%

Every figure computed from the deals behind it, none typed. Portfolio £50,633,699 • £8,000,000 uninvested cash

Deal Blotter
Live positions and active deals across the book

COUNTERPARTY	INSTRUMENT	PRINCIPAL	MEASURED	STAGE	FLAG
Meridian Bank plc 4.20% • 3 months	Deposit	£4,000,000	£4,009,205 principal plus 9,205 accrued over 20 days	awaiting confirmation	---

Northern Treasury Services Ltd
The lifecycle, and the checks it passed

MT320 NTS-2026-88213. Matched, and every field agreed.

- Maturity due**
The expected amount falls due. 5 Jul 2027
- Three way match**
The deal record, the confirmation and the statement. —
- Deal closed**
Headroom returns to the book. Until a deal closes the counterparty holds it, and the next deal may be blocked for no reason. —

THE CHECKS IT PASSED

This position predates the gate. Seeded deals carry no check run, which is why the seed is illustrative rather than evidence.

THE COUNTERPARTY'S RECORD
It arrives on its own route, which is the only reason matching is a control. Fold it into the deal feed and there is nothing to match against.

MT320 NTS-2026-88213
Every field agreed. MATCHED

AMENDMENTS
Nothing amended. A correction is an entry rather than an edit: the original accrual rows stay and reversals are written against them. [Raise one](#)

SETTLEMENT
The deal record, the counterparty's confirmation and the bank statement. Two of three agreeing is not enough: if the record and the confirmation agree but the statement differs, the money did not arrive as promised. [Match three sources](#)

Not settled. Until this deal closes the counterparty still holds the headroom, and the next deal may be blocked for no reason.

WHAT IF THIS DEAL IS NOT ROLLED AT MATURITY?
Matures 5 Jul 2027. Runs the arithmetic as if the cash lands in the operating account instead of being rolled. Nothing is written. [See what would happen](#)

WHAT WOULD HAPPEN
£18,119,836 lands in the operating account (£18,000,000 principal, £119,836 accrued). Northern Treasury Services Ltd used drops by £18,000,000. headroom rises by the same. Roughly £1.997 *drafted just now*

Figure 4. What if a deal is not rolled at maturity.

What it solves

"If we let this deposit run off, what happens?" is a routine question and the answer has three parts: how much cash lands, how much headroom returns to the counterparty, and how much daily interest stops accruing. The treasurer knows the shape but not the numbers, and looking them up interrupts a decision.

What it adds

- Sits on the deal panel itself, next to the maturity date, so the button is where the question would be asked.
- The paragraph names the cash arrival, the counterparty's used-and-headroom change, and the daily interest that would stop.
- Every figure is derived from the deal's stored accrual and the current book — no interpretation of what "not rolled" means; the maturity date is what the deal itself carries.

How it stays safe

Read-only, no scratch copy required. The maths is pure arithmetic on the current state.

Feature 05 · Cash Deployment Planner — four options, ranked

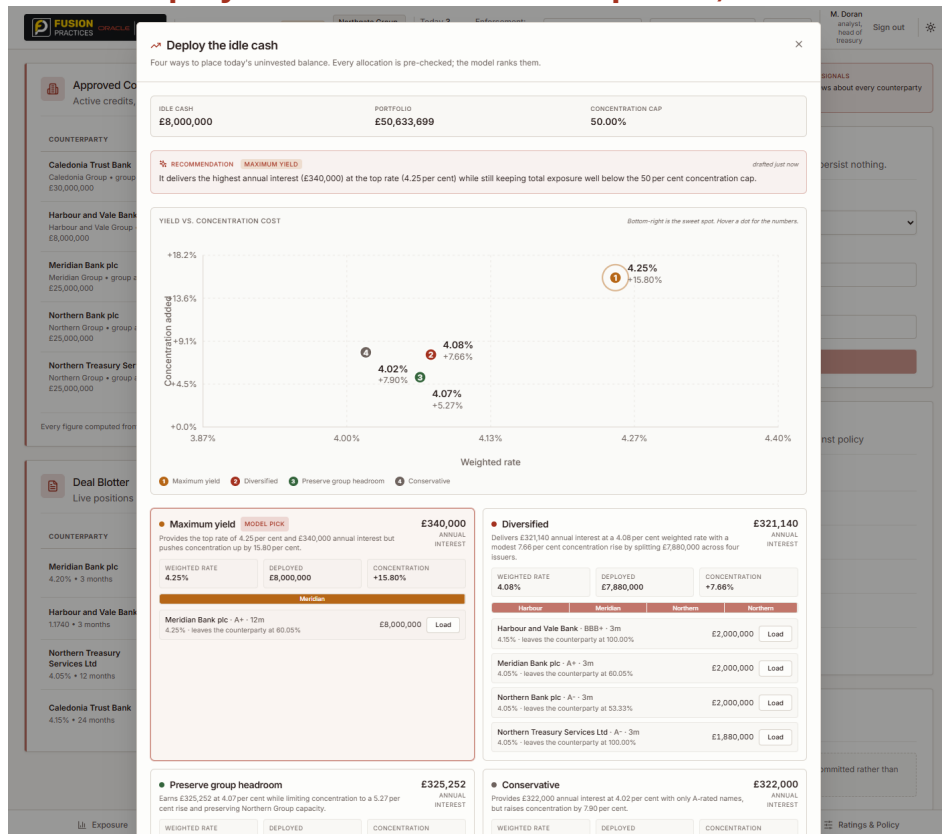


Figure 5. Cash Deployment Planner — four options, ranked.

What it solves

Placing idle cash well is a five-way trade-off — yield vs concentration vs group headroom vs maturity ladder vs credit quality. Nobody does that on paper. In practice a treasurer either picks the easiest counterparty or spends an hour in Excel each week. This is the strongest USP of the AI layer: it makes the choice explicit, side by side, with the trade-off visible.

What it adds

- One click produces four candidate plans, each already checked against the six checks — every candidate is bookable.
- The four archetypes cover the space: Maximum yield (concentrate on the highest-paying name), Diversified (split across four names at short tenor), Preserve group headroom (skip the tightest group), Conservative (A and above only).
- A yield-vs-concentration scatter chart places all four side by side so the trade-off is visible: bottom-right is the sweet spot.
- The AI recommends one candidate with reasoning, and writes a per-card trade-off label that quotes the real figures.
- One-click Load pipes the chosen allocation into the ticket, where the six checks re-run and the treasurer signs the same way as any other deal.

How it stays safe

Every allocation is pre-checked. Loading a candidate into the ticket runs the six checks again from scratch. The planner proposes; the ordinary deal path disposes.

Feature 06 · The candidate cards under the chart

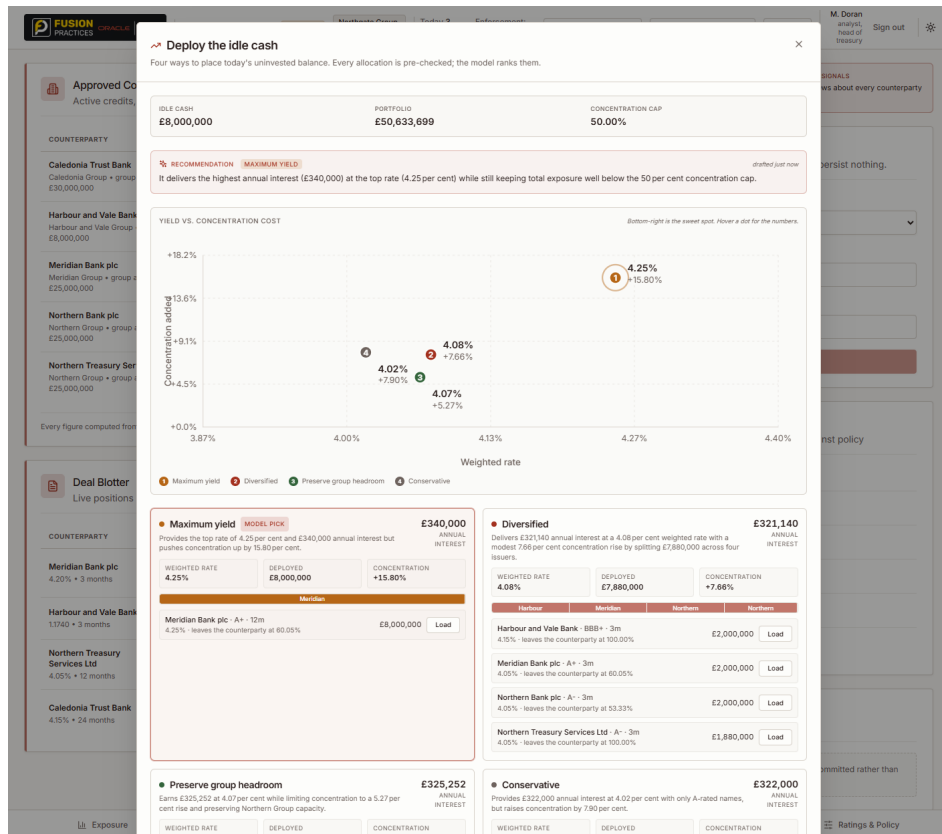


Figure 6. The candidate cards under the chart.

What it solves

The scatter chart tells the shape of the choice; the cards tell the detail. Each card shows the plan's allocations, counterparties, tenors, resulting utilisations — the whole thing a treasurer needs to check before loading.

What it adds

- Per-card allocation bar, colour-coded to the dot on the chart above, so eye can trace card to dot in one glance.
- Per-allocation row shows the counterparty, rating, tenor, rate, principal, and where it would leave the counterparty after the deal. A treasurer can approve or reject at the allocation level, not just the candidate level.
- The AI's per-card label sits at the top: "Deploys the full £8m at 4.25% but concentrates exposure to one A+ bank" — the trade-off, in one sentence.
- Model pick chip on the recommended card, so the AI's choice is unmistakable but never automatic.

How it stays safe

Every figure on every card is deterministic. The Load button pipes exactly those values into the ticket for the standard checks and signature.

Feature 07 · Ingest a confirmation from any format

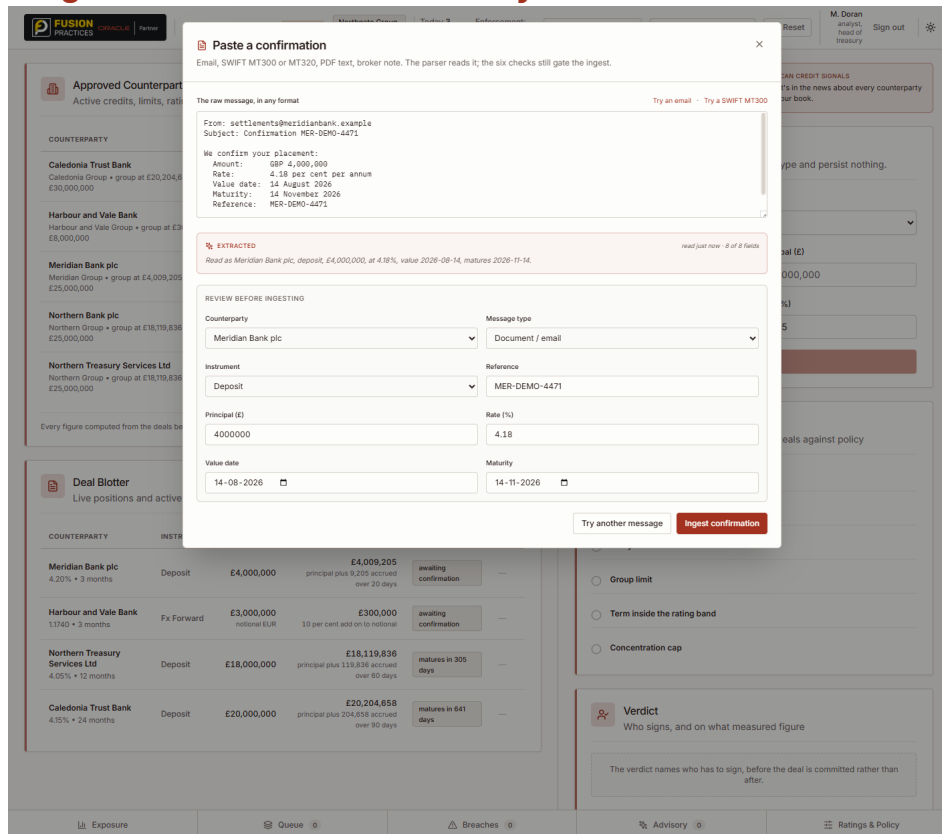


Figure 7. Ingest a confirmation from any format.

What it solves

Confirmations arrive from counterparties in every possible shape — SWIFT MT300s, MT320s, email bodies, PDF attachments, broker notes. The register's confirmation feed accepts JSON. In production this gap gets bridged by an operations team retyping every confirmation, which is where keying errors are born. This feature closes the gap.

What it adds

- Paste the raw message — email body, SWIFT text, PDF copy-paste — and the AI extracts the eight fields the matcher needs.
- The extracted fields are shown for review before ingest, with a summary line "Read as Meridian Bank plc, deposit, £4,000,000, at 4.18%, value 2026-08-14".
- Handles counterparty name matching in a case-insensitive way, so a SWIFT message with the name in ALL CAPS resolves to the canonical name on the book.
- Distinguishes an FX forward rate (1.1740) from a deposit rate (4.18%) automatically — both live in the same field on the model, and the AI reads the instrument correctly.
- Flags anything suspicious as a warning, so the reader sees "Maturity is before value date — one of them is wrong" before committing.

How it stays safe

The parser reads; the matcher decides. Nothing lands on the book from the parser alone. Every field is presented for review, and ingest goes through the standard confirmation endpoint and the standard matcher.

Feature 08 · Credit signals — news for every counterparty on the book

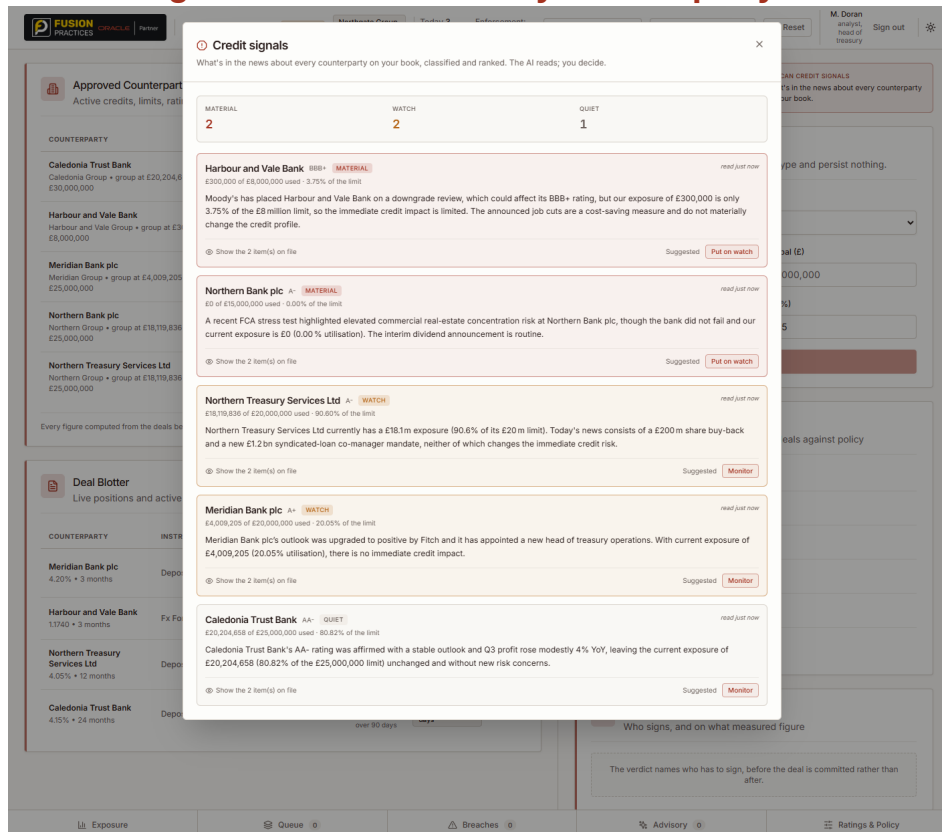


Figure 8. Credit signals — news for every counterparty on the book.

What it solves

A treasurer cannot personally read the FT for forty counterparties every morning. Which is why the biggest credit signal usually arrives via a headline someone else spotted — sometimes days late. The credit signal scanner reads for the treasurer, classifies every item, and ranks the counterparties by whether they need attention today.

What it adds

- One scan reads every news item on file for every counterparty. Classifies each one as MATERIAL, WATCH, or IMMATERIAL.
- Writes a summary line grounded in today's exposure. Not just "Moody's placed Harbour on review" but "Moody's placed Harbour on review, and we hold £300,000 of an £8m limit — 3.75% utilisation, low direct impact". The context is the register's own state.
- Suggests one action from a fixed set: Put on watch, Reduce, Roll off, Monitor, Ignore. Counters sorted by severity so the material ones surface first.
- Header stats: Material 2, Watch 1, Quiet 2 — so a morning scan is a glance at the numbers, not a scroll through everything.

How it stays safe

The scanner writes cards; it does not act. Every suggested action requires the treasurer to click. Any action taken goes through the ordinary Ratings, Queue or deal-panel controls.

Feature 09 · The signal card, expanded

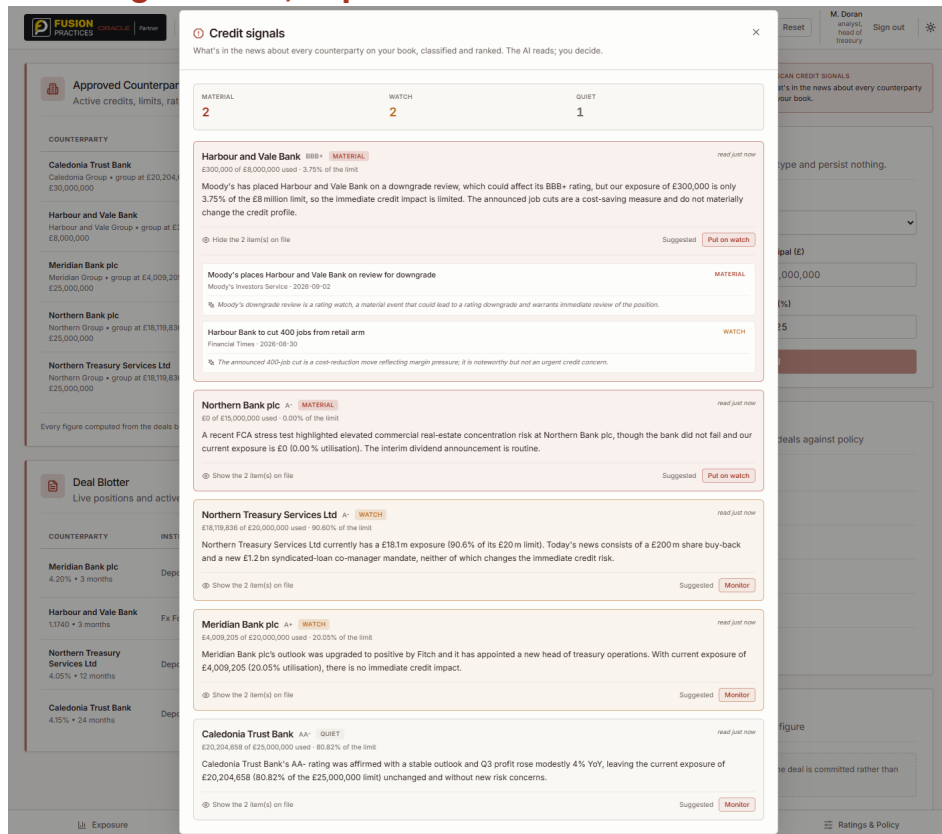


Figure 9. The signal card, expanded.

What it solves

A summary of "three items on file, material" is useful, but the auditable version is the item-by-item classification with the reasoning behind it. Expanding the card shows every news item and how the AI read it.

What it adds

- One row per underlying news item, with source, date, and the MATERIAL / WATCH / IMMATERIAL tag.
- The AI's per-item reasoning line under each headline, so a reader can see why an earnings miss was flagged as WATCH and a routine dividend as IMMATERIAL.
- Sparkle icon on the reasoning line makes it obvious which part is the AI's interpretation and which part is the raw source.

How it stays safe

The classifications are the AI's; the news items are the raw source. Both are shown, so a treasurer can override an IMMATERIAL classification by clicking Put on watch anyway.

Feature 10 · The reasoning trace

Treasury Register · Northgate Group Treasury · Today 3 Sept 2026

Approved Counterparties

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Meridian Bank plc Meridian Group • group at £4,009,205 of £25,000,000	A+	£20,000,000 to 12 months	£4,009,205	£15,990,795 free 20%
Northern Bank plc Northern Group • group at £18,119,836 of £25,000,000	A-	£15,000,000 to 6 months	£0	£15,000,000 free 0%
Northern Treasury Services Ltd Northern Group • group at £18,119,836 of £25,000,000	A-	£20,000,000 to 12 months	£18,119,836	£1,880,164 free 91%

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Exposure Queue Breaches

Ratings and policy

The world moving, and the two settings that are still open

RECORD A RATING ACTION

Counterparty: Caledonia Trust Bank — AA-

New rating: BBB

Caledonia Trust Bank holds £20,204,658 against £25,000,000. At BBB the limit becomes £5,000,000 to 3 months, and every live position is re-tested against it.

% Thinking... **Apply, and re-test the book**

REASONING

- ✓ Taking a snapshot of the current book.
- ✓ Applying the rating action to a scratch copy.
- ✓ Tightening the limit to the new band's ceiling.
- ✓ Re-testing every position that names this counterparty.

ENFORCEMENT

Both are built as a policy setting, which takes the decision off the critical path. Settle it by flipping it on a failing deal in the room rather than in a slide. Changing it writes a new policy version, so runs recorded under the old one stay reproducible.

Hard block Warn with an override

THE CLOCK

Everything derives from it. Moving it forward accrues interest on every deposit, which moves every measured exposure with it.

03-09-2026

Figure 10. The reasoning trace.

What it solves

AI features arrive with a network call, and a spinner during that call gives the reader no idea what is actually being done. Worse: it reads as generic, which is exactly what an AI response should not read as. The trace fixes this by showing the reader what the system is doing in the vocabulary of their own work.

What it adds

- One step at a time, ticking through with a green check as each completes.
- Every step corresponds to real server-side work — not invented. "Taking a snapshot of the current book", "Applying the rating action to a scratch copy", "Rolling the scratch copy back" are what the code actually does.
- Held until every step has been shown, so a fast Groq response doesn't flash past the reasoning — the reader always sees the sequence.

How it stays safe

The trace is client-side pacing, not fake work. The step text must match the code; the discipline that keeps it honest is that changing the code without updating the trace makes the trace a lie, and the whole feature loses its credibility.

The whole AI layer, in one summary

Feature	Real problem it solves	How it stays safe
Advisory ranking	Where to place today's cash, ranked	Only ranks pre-checked candidates
What if a rating changes	How a downgrade would move the book	Runs in a savepoint, rolled back
What if the cap moved	Which groups would breach a tighter cap	Read-only, no writes
What if a deal is not rolled	Cash landing, headroom, income foregone	Read-only arithmetic
Cash Deployment Planner	Four bookable ways to place idle cash	Every allocation pre-checked, six checks re-run on Load
Confirmation parser	Reads any confirmation format	Fields shown for review, ingest via standard endpoint
Credit signal scanner	Reads news for every counterparty	Writes cards; every action requires the treasurer to click
Reasoning trace	Makes the AI's work visible	Steps match the real server actions

The one line worth remembering

The AI does not write anything to the book. The six checks gate every path in, whether the deal came from a keystroke, a corrected confirmation, an amendment, or the planner. If the model disappeared tomorrow, every feature would fall back to a deterministic paragraph and the app would still work — slower to read, but nothing about safety would change.

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